

0.1 RQ5-Specific Coding Codebook

Coding purpose. RQ5 examines how the included reviews treat validation practice, which validation modalities they make visible, how rigorously they discuss the transfer of controllers from simulation to real buildings, and which validation gaps they signal.

Terminology. One term is used throughout for the mismatch between simulated and real building environments: the simulation-to-reality gap. Where individual reviews use other wording, such as sim-to-real transfer or deployment readiness, the audit tables record the review’s own phrasing and the coding maps it to this construct. Deployment readiness is treated as a separate construct, covering what a review says about preparing and operating a controller in a real building, rather than as a synonym.

Dimension naming. The dimension recorded in earlier drafts as validation-maturity treatment is named validation-practice treatment here. The change is terminological. It alters no score and no coding decision.

What the scale measures. The shared 0–3 scale records treatment intensity, meaning how prominently a review treats a coded modality, rigor element, or gap within its own account of the literature. It does not measure how much real-building evidence exists, and it does not measure whether the validation reported in the primary studies was sound. A high score on the real-building modality means the review makes field deployment visible, not that field deployment is widespread. The score describes the review, not the field.

Table 1: Shared treatment-intensity scale as applied in RQ5.

Score	Label	General interpretation
0	Not addressed	The modality, rigor element, or gap is not identifiable in the review.
1	Mentioned	It is acknowledged in passing, for example as a general statement that deployment remains limited, without further development.
2	Discussed	It is explained and used in the review’s discussion, but does not organize part of the synthesis.
3	Structuring role	It organizes part of the review’s synthesis, through a taxonomy, an explicit comparative criterion, a dedicated section, or a recurring interpretive theme.

Structure of RQ5 coding. RQ5 records four survey-level scores and three unit-level tables.

Validation-practice treatment records how developed the review’s overall treatment of validation practice is, ranging from a passing statement that deployment is limited to a structured account of how thoroughly reviewed applications have been tested. It is coded directly and has no single unit counterpart.

Deployment-readiness treatment records what the review says about preparing and operating a controller in a real building, including pretraining, online relearning, recalibration, safety checks, and fallback control.

Simulation-to-reality gap treatment records how rigorously the review discusses the mismatch between simulated training environments and real buildings, including simulator fidelity, pretraining, and transfer.

Gap-signal visibility records how far the review signals unresolved validation gaps. It is coded directly.

Validation modality has five units recording how visibly the review treats each form of evidence: simulation; emulation or testbed; pilot or demonstration; real building or field deployment; and cross-site, transfer, or generalization evidence.

Validation rigor has five units: validation taxonomy clarity; deployment-readiness discussion; simulation-to-reality gap rigor; safety or safeguards for deployment; and strength of real-world evidence.

Gap signals has five units: overreliance on simulation; absence of pilots or field tests; weak simulation-to-reality methodology; limited deployment protocol detail; and limited long-term real-world evidence.

Relationship between survey-level and unit-level scores. Two survey-level scores are the same coded value as a rigor unit, recorded once and reported in both places. Deployment-readiness treatment corresponds to the deployment-readiness discussion unit, and simulation-to-reality gap treatment corresponds to the simulation-to-reality gap rigor unit. Any aggregate computed across both tables should count each once.

Validation-practice treatment and gap-signal visibility are coded directly from the review and are not derived from unit scores. Each coincides with the highest score in the corresponding unit table in some reviews but not in others, so neither is an aggregation. A review can classify one modality in detail while its overall treatment of validation practice remains undeveloped.

Zero and applicability. A score of 0 means the review does not address the modality, rigor element, or gap. The scale does not separate a topic that a review considered and did not develop from one that fell outside its declared scope. Both are recorded as 0. This keeps denominators uniform and is consistent with the construct, which records the treatment a review gives a topic rather than the reasons for that treatment. A score of 0 should therefore not be read as evidence that a review overlooked something it could have covered.

All coded reviews are scored on every dimension and unit. RQ5 has no applicability route, so the denominator is the full coded corpus.

Table 2: Operational criteria for the survey-level validation-practice score.

Score	Criterion
0	How reviewed work was validated is not addressed.
1	The review states that validation or deployment remains limited or early-stage, without developing the point.
2	The review discusses validation practice substantively, for example through simulator dependence, environment-interaction burden, or the cost of physical exploration, but does not construct a staged validation account.
3	The review organizes part of its synthesis around validation practice, for example by summarizing deployment settings and modes, or by reporting how thoroughly reviewed applications have been tested.

Operational criteria: validation-practice treatment.

Table 3: Operational criteria for validation-modality units.

Score	Criterion
0	The modality is not identifiable in the review.
1	The modality is named or noted in passing, for example a remark that few studies reach real buildings.
2	The modality is discussed substantively, with examples or an account of what it involves.
3	The modality is used as a classification criterion across reviewed studies, or organizes part of the synthesis.

Operational criteria: validation modality.

Table 4: Operational criteria for validation-rigor units.

Score	Criterion
0	The element is not addressed.
1	The element is acknowledged without development, for example a brief distinction between offline training and online interaction.
2	The element is explained, with its implications for validation made clear.
3	The element is a structuring concern, developed through a dedicated account, an explicit framework, or a recurring basis for the review’s conclusions.

Operational criteria: validation rigor.

Table 5: Operational criteria for gap-signal units and for the survey-level gap-signal score.

Score	Criterion
0	The gap is not signaled.
1	The gap is acknowledged as a limitation without elaboration.
2	The gap is explained, with its consequence for validation or deployment made clear.
3	The gap is a structuring concern, developed as a recurring theme, a dedicated discussion, or an explicit basis for the review’s recommendations.

Operational criteria: gap signals.

Evidence rules. Tables, figures, taxonomies, and dedicated sections are coding evidence, and organizing use of a modality, rigor element, or gap in any of them supports a structuring score. A general statement that deployment remains limited supports a score of 1 and does not on its own support a higher score for any specific modality. Repeated mentions do not accumulate into a higher score. Content appearing in the abstract or keywords alone is scored at most 1. Supplementary material was not coded, because its availability is inconsistent across the corpus and coding it would make reviews non-comparable. Coding reflects the review authors’ own account: a field deployment reported in a cited primary study is insufficient unless the review discusses, tabulates, or classifies it. Where evidence falls between two adjacent levels, the lower score is recorded unless the higher criterion is clearly satisfied.

Worked boundary examples. The examples below are taken from the survey-level audit tables that follow.

Summary fields. Each survey-level audit table also carries three narrative fields: validation orientation, dominant validation framing, and the main RQ5 consequence. These summarize the coded scores in prose and are not themselves coded values.

Table 6: Worked boundary examples for RQ5 coding.

Review	Dimension or unit	Score	Basis for the boundary
Al-Ani and Das (2022)	Validation practice, survey	1	The review states that RL in home energy management is early-stage and that deployment remains limited, but validation does not structure the review. Boundary against 2.
Mason and Grijalva (2019)	Validation practice, survey	2	Validation is discussed through simulator dependence, the cost of physical exploration, and the need for representative data, without a staged validation account. Boundary against both 1 and 3.
Weinberg et al. (2023)	Validation practice, survey	2	Practical readiness is discussed through sample efficiency, transfer, scalability, and risky exploration, but no staged validation account is built. Contrast with deployment readiness in the same review, scored 3.
Kannari et al. (2025)	Validation practice, survey	3	The review summarizes practical deployments, implementation settings, deployment modes, and how thoroughly existing applications have been tested.
Al-Ani and Das (2022)	Simulation-to-reality gap	1	The review notes the use of models and simulation-based training, without sustained analysis of transfer to real buildings. Boundary against 2.
Weinberg et al. (2023)	Simulation-to-reality gap	3	Transferability, sample efficiency, environment-interaction burden, and model-based RL are central to how the review frames practical use.
Mason and Grijalva (2019)	Real building, modality	1	Physical implementation is noted as a later refinement step after simulation pretraining, without substantive discussion of field evidence. Boundary against 2.
Kannari et al. (2025)	Real building, modality	3	Real-building deployment is the review’s classification axis, organizing its account of implementation settings and deployment modes.